Scalability & Future Plan

Date	02 July 2024
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine.
Maximum Marks	1 Mark

Scalability & Future Plan:

The current **OptiCrop** system is designed as a web-based crop recommendation platform that can be easily scaled to support a larger number of users and additional functionalities in the future.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of users on a local server.	Deploy on cloud platforms such as Render, AWS, or Azure with load balancing.
2	Data Storage	Uses local CSV dataset and model files.	Integrate MySQL or MongoDB for large-scale agricultural data storage.
3	Performance	Predictions are generated using a locally stored model.	Optimize model performance and use caching techniques for faster predictions.

4	Security	Basic input validation is implemented.	Add user authentication, HTTPS, and secure API integration.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only a limited agricultural dataset.	May reduce prediction accuracy for other regions and crops.	High
2	Application is currently designed for local deployment only.	Cannot support a large number of simultaneous users.	Medium
3	No user authentication and cloud database integration.	Limited security and scalability.	Medium

Scalability Plan:

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment and online hosting	2 Months	Improve accessibility and support more users.
Phase 2	Real-time weather API integration	3 Months	Provide more accurate crop recommendations.
Phase 3	Mobile application development	4 Months	Enable farmers to access the system through smartphones.

Phase 4	Advanced AI models and multilingual support	6 Months	Increase prediction accuracy and usability for farmers in different regions.